

Machine Learning Lab – Question 7

Question

Write a program to implement Logistic Regression (LR) algorithm in Python.

Code

```
from sklearn.datasets import load_iris
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import accuracy_score

iris = load_iris()
X_train, X_test, y_train, y_test = train_test_split(
    iris.data, iris.target, test_size=0.2, random_state=42, stratify=iris.target)

model = LogisticRegression(max_iter=1000)
model.fit(X_train, y_train)
pred = model.predict(X_test)

print("Accuracy:", round(accuracy_score(y_test, pred), 4))
print("Sample prediction:", iris.target_names[model.predict([[5.1, 3.5, 1.4, 0.2]])[0]])
```

Output

```
Accuracy: 0.9667
Sample prediction: setosa
```